

Causal Operator–Operand Factorization in the Residual Stream of LLMs

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Abstract

We study how open-weights transformers (Qwen3-1.7B/8B, Gemma-2-9B) represent a **relational operation** applied to an entity — *currency-of*, *capital-of*, *language-of*, *demonym-of*, *continent-of* applied to a country. We find that the operation is carried by a **manipulable operator direction** in the residual stream that is largely **separable from the operand**. Adding $v(\text{op_B}) - v(\text{op_A})$ to a prompt performing op_A converts the model's answer to op_B for all 20 ordered operator pairs (matched-norm random controls null), and the direction **generalizes to operands never used to build it**, to **unseen paraphrase frames** (every build→test frame combination flips), and **across architectures** — the full pipeline replicates on Gemma-2-9B with the largest effect sizes of the three models tested. A two-way analysis of variance factorizes the representation as $H[\text{operand}, \text{operator}] = \mu + \text{operand} + \text{operator} + \text{interaction}$, ~90% additive with a ~10% **fusion** interaction, operator and operand occupying largely orthogonal subspaces; the operation-dominant representation *emerges along the sequence* (operand-dominant at the entity token → operator-dominant at the query token). Strikingly, the operation is **separable from its surface realization**: *language-of* and *demonym-of* are distinct operator directions even though both emit "Italian". As an explanatory device we borrow a morphological metaphor — the model **declines** a concept by its functional role (case), with **syncretism** (shared surface form) and **fusion** (non-additive stem+ending) as in a Latin declension. Testing generalization across domains, the factorization is **specific to relational retrieval**: for **arithmetic** (+, ×, -) and **comparison-logic** operators, the interaction term is 2–4× larger and the operator **fails to generalize to held-out operands**, consistent with arithmetic being

computed by a "bag of heuristics" rather than a linear operator. A control that instead varies a *linear numeric parameter* (add-N) rather than the function generalizes better and is near-collinear, reconciling a contrary recent report (Christ et al., 2025). Finally, the structure is **causal, not a readout advantage**: the Jacobian-lens ("J-space") readout is not more legible than the raw residual on any of our metrics under matched controls.

1 Introduction

A line of work shows that a *task* or *relation* can be captured by a single addable vector in an LLM's residual stream (task vectors, Hendel et al., 2023; function vectors, Todd et al., 2024) and that many relations are approximately linear operators (LRE, Hernandez et al., 2024; Merullo et al., 2024). What has not been characterized is the **joint structure of operator and operand**: whether the operation *factorizes* from its argument, whether that factorization *generalizes*, whether the operation is separable from the *word it produces*, and whether any of this holds *beyond* relational facts, in arithmetic and logic. We answer these on Qwen3, and offer a morphological reading — relational computation as **declension** — as an organizing metaphor that anticipates the specific asymmetries we observe (syncretism at the level of the surface form, fusion in the interaction term).

2 Related work and what is new here

Operation as an addable vector. Task Vectors (Hendel et al., 2023) and Function Vectors (Todd et al., 2024) extract one vector per ICL task and add/patch it to trigger the task. In-Context Vectors (Liu et al., 2024) and ActAdd (Turner et al., 2023) steer a task/style similarly. These bundle operator *and* operand into one unfactored "task" direction and (for FVs) show *partial* cross-task arithmetic without null controls. We contribute the **factorization** (operator $\oplus$ operand), an **exhaustive**

all-pairs swap with matched-norm nulls, and a **held-out-operand generalization** test.

Relation as a linear operator. LRE (Hernandez et al., 2024) fits a full affine map $W \cdot s + b$ per relation. That represents the relation as a *monolithic* operator; we show a **low-dimensional additive operator direction** that is arithmetically composable and **factorable from the operand**, and we quantify the operand/operator/interaction variance split, which LRE does not.

Additive factual recall. Summing Up the Facts (Chughtai et al., 2024) decomposes factual recall into additive *circuits* (subject vs relation contributions). Ours is a **representational two-way ANOVA of the residual state** with an explicit, measured **interaction/fusion term** — a different object.

Attribute subspaces / reading position. RAVEL (Huang et al., 2024) disentangles multiple *attributes of one entity* into subspaces (operators sharing an operand); we measure the orthogonal axis, **operator vs operand**. The operand $\rightarrow$ operator shift along the sequence restates the subject-enrichment $\rightarrow$ attribute-extraction dynamics of Geva et al. (2023) in operator/operand terms — our least novel point, included as confirmation.

Arithmetic and logic. Arithmetic in LLMs is reported as a "bag of heuristics" (Nikankin et al., 2025) and Fourier/helical procedures (Nanda et al., 2023; Zhong et al., 2023; Kantamneni and Tegmark, 2025), with the operand *value* cleanly linear (Gurnee and Tegmark, 2024). Truth is a steerable direction (Marks and Tegmark, 2023) and logical-operator heads exist (Hong et al., 2024). No prior work reports a **cross-domain operator/operand factorization**; we provide one and show it **breaks down** for arithmetic and logic — itself a result.

What is new here is the combination, not any single ingredient. That relations admit addable vectors (task/function vectors) and linear decoders (LRE) is established; what has not been done is the joint, quantified analysis: (1) an operator $\oplus$ operand **factorization of the residual state itself** (two-way ANOVA with a measured interaction/fusion term and operator/operand subspace angles); (2) an **exhaustive all-pairs causal swap** with matched-norm nulls; (3) **held-out-operand generalization** as the discriminator between a real operator and interpolation among build examples; (4) the **operation-vs-realization disso-**

ciation (language $\neq$ demonym as directions despite identical output); and (5) a **cross-domain test** showing the factorization is specific to relational retrieval and breaks down for arithmetic and logic. The declension/case reading is our *presentation* of these measurements — an organizing metaphor that anticipates the observed asymmetries (syncretism at the exponent, fusion in the interaction term) — not an additional empirical claim.

3 Method

Setup. Qwen3-1.7B/8B throughout, plus Gemma-2-9B for the cross-architecture replication (§4.5); all run in bf16 on a single AMD Strix Halo APU (see setup). A relation is rendered into a template; the canonical frame is "The {op} of {a} is", and §4.4 additionally uses two paraphrase frames (question-answer and discourse-prefixed) that hold the {op} of {a} unit fixed while varying the surrounding frame, all ending in *is* so answer scoring is comparable. Reading position is the query token unless stated; the depth "workspace" band (38–92% of layers) is where Gurnee et al. (2026) locates the global workspace and where we build operator directions.

Operator direction. For operator k , $v(k) = \text{mean_operand}[h(\text{operand}, k) - \text{mean_op } h(\text{operand}, \cdot)]$ at the workspace layers — the deviation of an operator from the operand's average over operators, averaged over operands (the function-vector construction; `scripts/op_core.py:op_dirs`).

Swap and controls. Efficacy of $k_A \rightarrow k_B$ is the change in $\text{logit}(\text{answer}_B) - \text{logit}(\text{answer}_A)$ at the query position when $\alpha \cdot (v(k_B) - v(k_A))$ is added over the band, versus a **matched-norm random** direction. Answers are scored by their distinguishing token (bare digit for arithmetic, since Qwen3 splits "3" into [space, 3]); a tokenization guard drops operator pairs that collide at that token.

Factorization. $H[\text{operand}, \text{operator}] = \mu + \text{operand}(o) + \text{operator}(k) + \text{interaction}$; we report the variance share of each term and the principal angles between the operand- and operator-subspaces (`scripts/op_core.py:factorize`).

Generalization. Build $v(\text{op})$ from half the operands; test the swap on the held-out half. This distinguishes a genuine operator from interpola-

tion among the examples used to build it.

Statistical treatment. The 20 ordered swaps are **not** 20 independent observations: they are built from 5 operator directions (each participating in 8 ordered pairs) and evaluated on the same 12 operands. We therefore report cluster-bootstrap percentile intervals (10,000 replicates) at two levels. Within a pair, the **operand** is the resampling unit (per-pair 95% CIs). Across the paradigm, the **operator** is the top-level cluster: a dyadic node bootstrap resamples the operator set with replacement, weights each ordered pair by the product of its endpoints' multiplicities, and resamples operands within surviving pairs (`op_core.bootstrap_pair_ci / bootstrap_family_ci`; per-operand values persisted in the `*_long.parquet` artifacts).

J-space controls. We repeat readout-geometry in the J-lens readout `unembed(J·h)` vs the logit-lens readout `unembed(h)`, and re-run efficacy with **spectrum-matched random-projection** and **permuted-vocabulary** null lenses (`scripts/control_lens.py`).

4 Results

4.1 Relational operators are manipulable directions

All 20 ordered operator swaps flip the answer (mean swap **+21** logit units on 1.7B, **+25** on 8B) while the matched-norm random control is ~ 0 . Representative (1.7B):

swap	clean	+operator	+random
currency $\rightarrow$ capital	-7.9	+31.5	-1.2
capital $\rightarrow$ currency	-12.0	+33.0	-2.5
capital $\rightarrow$ language	-8.5	+32.5	+3.0
continent $\rightarrow$ language	-1.9	+23.0	+1.4

Under the operator-level cluster bootstrap (§3, the level that respects that pairs share directions), the swap-random contrast is **+22.6, 95% CI [+14.0, +32.1]** at 1.7B and **+26.0 [+17.9, +32.8]** at 8B; the flip fraction is **1.00 [1.00, 1.00]** at both scales — every replicate flips every pair. Per-pair operand-bootstrap CIs never cross zero:

4.2 The representation factorizes into operator $\oplus$ operand

Two-way ANOVA at a mid-workspace layer (1.7B / 8B):

$\sim 90\%$ additive; operand/operator subspaces largely orthogonal (principal angles 41–82°).

The operation- dominant representation **emerges along the sequence**: the entity enters operand-dominant and is *declined* into an operator-marked form by the query position (the reading-position control rules out template echo; the causal swaps are the stronger control).

4.3 Operation $\neq$ realization (syncretism)

language-of and *demonym-of* emit the **same word** (Italian) yet are **distinct operator directions**; the swap between them is weak precisely because they share an exponent. A **pure desinence** built from the shared-output pairs (`mean[h(language) - h(demonym)]`, exponent cancelled) still installs the relation causally (1.7B: $-3.7 \rightarrow +12.5$). The syncretism appears at the *exponent* level, not the *case* level — as in Latin declension.

4.4 Operators generalize to held-out operands and across prompt frames

Held-out operands. `v(op)` built on 6 operands and applied to the 6 held-out operands still flips **20/20** swaps — a genuine, transferable operator, not interpolation. Operator-level cluster bootstrap on the held-out contrast: **+20.0 [+10.8, +29.5]** at 1.7B, **+22.8 [+14.0, +31.0]** at 8B; flip fraction **1.00 [1.00, 1.00]** at both.

Cross-frame transfer (paraphrase robustness). To rule out template echo we render every relation in three paraphrase frames — declarative (The {op} of {a} is), question–answer (Q: What is the {op} of {a}? A: It is), and discourse-prefixed (It is well known that the {op} of {a} is) — and test every build→test frame combination: `v(op)` built on frame *i*, all-pairs swap run on frame *j*. On 1.7B, **all 9 combinations flip 20/20** (180/180; off-diagonal transfer 120/120, mean contrast **+24.2**), and the contrast is *frame-invariant to two decimals*: directions built on the declarative frame produce +22.6 on all three test frames. The clean baselines do shift across frames ($-7.0 / -8.1 / -7.9$), confirming the frames are genuinely different prompts; the operator's causal effect does not. At 8B the pattern replicates — **100/100 flips over the tested combinations (80/80 cross-frame, mean contrast +28.7)**, with the declarative-built direction again frame-invariant ($+26.0 / +26.0 / +26.0$). The operator direction is a property of the **relation**, not of the prompt that elicits it.

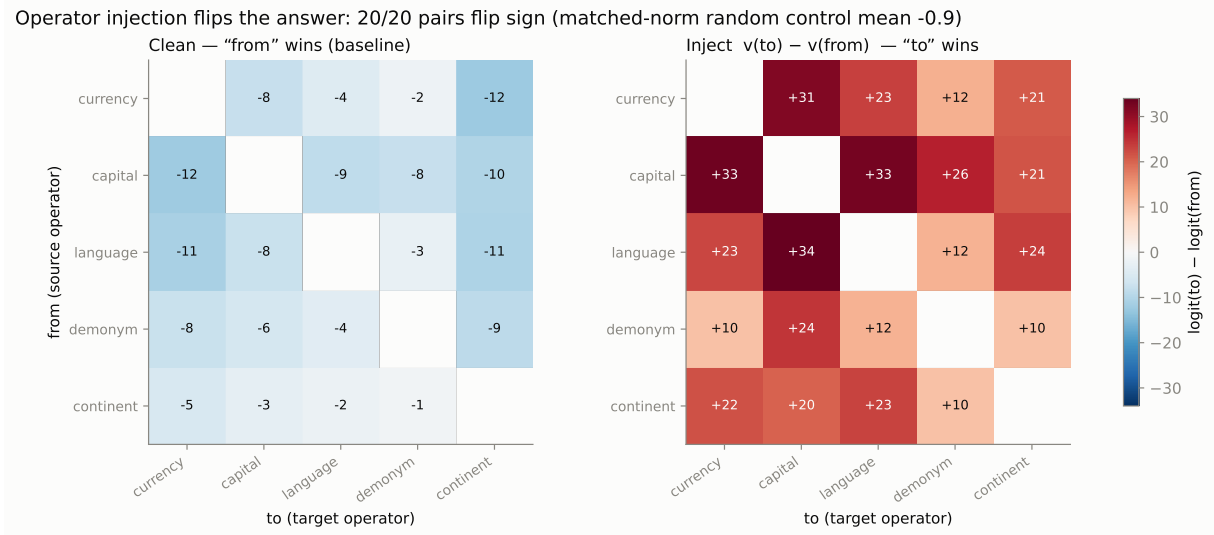


Figure 1: All-pairs operator swap: clean (baseline, "from" wins, blue) vs. injecting $v(\text{to}) - v(\text{from})$ ("to" wins, red). 20/20 pairs flip sign; matched-norm random control ≈ 0 .

read position	operand	operator	interaction (fusion)
query token	5% / 6%	86% / 82%	9% / 13%
entity token	59% / 55%	31% / 34%	9% / 11%

Table 1: Two-way ANOVA of the workspace state: variance shares by read position (1.7B / 8B).

4.5 Cross-architecture replication (Gemma-2-9B)

The entire pipeline transfers unchanged to **Gemma-2-9B** — a different pretraining corpus, tokenizer (SentencePiece; answer single-token coverage 0.95), and architecture family (soft-capped logits, GQA):

Every qualitative signature replicates — all-pairs flips with matched-norm nulls ≈ 0 , held-out-operand transfer, case-dominant factorization at the query token with single-digit fusion, the along-sequence shift (stem 37.6% $\rightarrow$ 8.4% from the entity token to the query token), and the exponent-free desinence. Effect sizes are, if anything, largest in Gemma. The operator–operand factorization is not an idiosyncrasy of one model family.

4.6 The factorization is domain-specific (arithmetic and logic do not)

Extending the identical pipeline to arithmetic (+, $\times$, -) and comparison-logic operators, on 1.7B:

The gradient replicates and sharpens at **8B**: relational held-out generalization 20/20 (interaction 13%); arithmetic interaction 25%, held-out 1/6; logic interaction 45%, held-out **0/6** (swap below random). Monotone at both scales: relational $\gg$ arithmetic > logic.

Relational operators factorize cleanly and generalize; arithmetic and logical operators are **entangled with their operands** (2–4 $\times$ the interaction) and do **not** form a transferable direction — consistent with arithmetic being a bag of heuristics / Fourier computation rather than a linear operator. The **held-out generalization test is the discriminator** between a real operator and memorized interpolation.

4.7 Reconciling with a contrary report: add-N vs. two-operand arithmetic

Christ et al. (2025) report the *opposite* sign for arithmetic: an operator built from an **add-N** relation (a fixed addend N applied to a single number) **does** generalize to held-out relations. The two results do not conflict — they cut arithmetic along different axes. In add-N the operator is **the addend N** over one numeric operand, so "generalizing across N" is interpolation along the number line (the operand *value* is itself linear; Gurnee and Tegmark, 2024), not transfer of a *function*. Our + $\times$ - cut varies the **function** over two operands. Running both cuts on Qwen3 (op_core.py --domain arith_addN vs arithmetic; scripts/operator_collinearity.py) places add-N exactly between relations and the

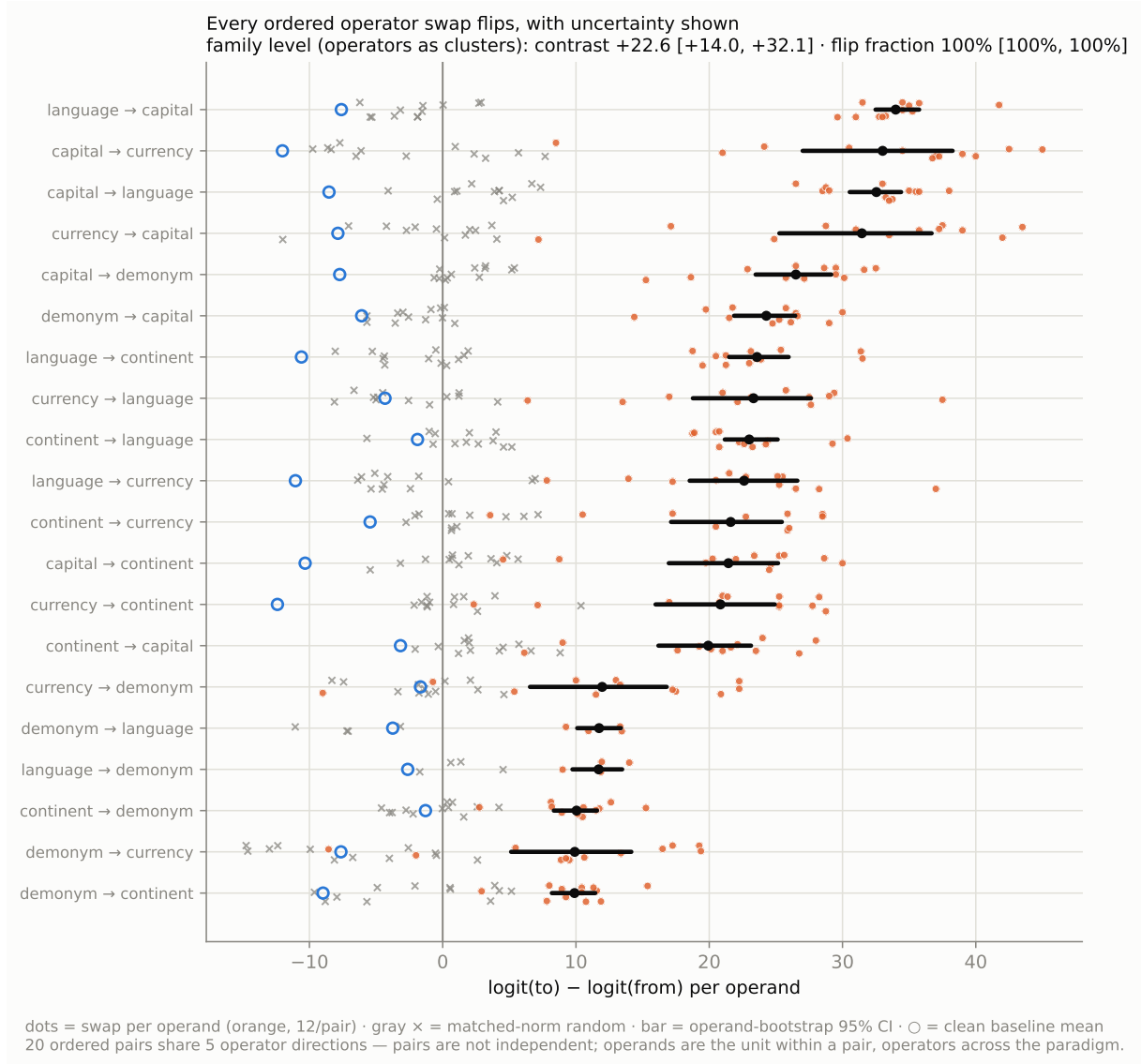


Figure 2: Per-pair distributions: swap values per operand with operand-bootstrap 95% CIs, matched-norm random control, and clean baselines. No CI crosses zero; the weakest pairs are exactly the syncretic ones (demonym↔language and their neighbours).

measure	Qwen3-1.7B	Qwen3-8B	Gemma-2-9B
all-pairs swap contrast	+22.6 [+14.0, +32.1]	+26.0 [+17.9, +32.8]	+30.4 [+23.8, +34.8]
flips (swap > 0)	20/20	20/20	20/20
held-out-operand contrast	+20.0 [+10.8, +29.5]	+22.8 [+14.0, +31.0]	+26.6 [+20.6, +33.2]
operator variance @ query	86%	82%	84.8%
interaction (fusion)	9%	13%	6.8%
pure desinence (clean → +v)	-3.7 → +12.5	-2.9 → +8.1	-3.0 → +8.6

Table 2: Cross-domain factorization: relational operators factorize and generalize; arithmetic and logic do not.

genuine-function cut:

add-N generalizes **better** than + × − (positive vs. clearly negative) and its operator directions are the **most collinear** of the three families — 76% of the operator-set variance on a single line — consistent with add-N being a linear numeric family rather than a set of distinct operations. This re-

produces Christ et al.'s positive (their cut varies a linear parameter) alongside our negative (ours varies the function) with no contradiction. The caveat is quantitative: Qwen3 BPE restricts us to single-digit results, so the add-N grid is small (5 operands) and its generalization is weak in absolute terms — the **ordering** (relations ≫ add-N >

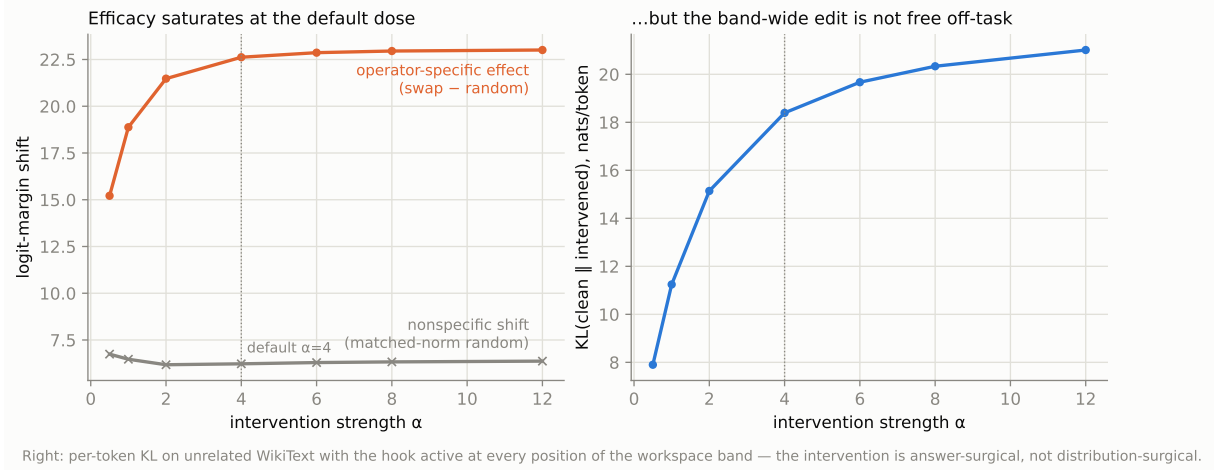


Figure 3: Dose-response: the operator-specific effect saturates at the default dose $\alpha=4$ while the nonspecific random shift stays flat; per-token KL on unrelated text grows monotonically — the edit is answer-surgical, not distribution-surgical.

domain	operator variance	interaction (fusion)	held-out generalization	swap vs random
relational	86%	9%	20/20 flip	+21 vs ~0
arithmetic	55%	23%	2/6 ($\leq$ random)	+0.4 vs +0.3
logic (compare)	33%	34%	2/6 ($\leq$ random)	+1.2 vs +0.2

Table 3: Cross-domain factorization: relational operators factorize and generalize; arithmetic and logic do not.

cut	held-out generalization (1.7B / 8B)	operator-set collinearity (top-1 var)
relations (5 operations)	20/20 · 20/20	0.39 (spread paradigm)
add-N (operator = addend)	5/12 · 6/12 (+0.10 / +0.05)	0.76 (most 1-D / number-line)
+ × − (operator = function)	2/6 · 1/6 (-0.43 / -0.55)	0.64

+ × −), not the magnitude, is the claim.

4.8 This is causal structure, not a J-space readout advantage

On four readout comparisons (bridge-entity pass@k, a surface-form logit difference, a number probe, and a concept-plane trajectory), the J-lens readout shows **no advantage** over the logit-lens under matched controls; a spectrum-matched random projection reads number as well as the fitted J-lens, and the concept-plane "channel" does not survive a random-plane baseline (see findings §Part 1). The operator/operand structure lives in the **causal** organization of the residual, and is **not** more legible in the J-space than in the raw stream (its 1.7B readout-geometry difference does not replicate at 8B). We therefore make no claim of a special readable subspace; the contribution is the operator/operand *causal* factorization.

5 Limitations

- **Scale and family.** 1.7–9B models from two families (Qwen3, Gemma 2) — both decoder-only transformers; no ≥ 30 B model. Relational linearity holds for a subset of relations even in-domain (cf. LRE ~48%).
- **Relation coverage.** Five hand-chosen country relations, English-only. Prompt variation covers three paraphrase *frames* (§4.4) but all frames share the {op} of {a} phrasing unit; phrase-level paraphrases, more relations, and other entity types remain open.
- **Arithmetic coverage.** Qwen3 BPE splits multi-digit numbers, so we restrict to single-digit results; the arithmetic null is on that regime and at 1.7B/8B, where arithmetic competence is itself limited.
- **Reading-position result** restates known subject-enrichment→attribute-extraction dynamics.
- **No J-space claim.** The readout-geometry difference is 1.7B-specific; the paper stands

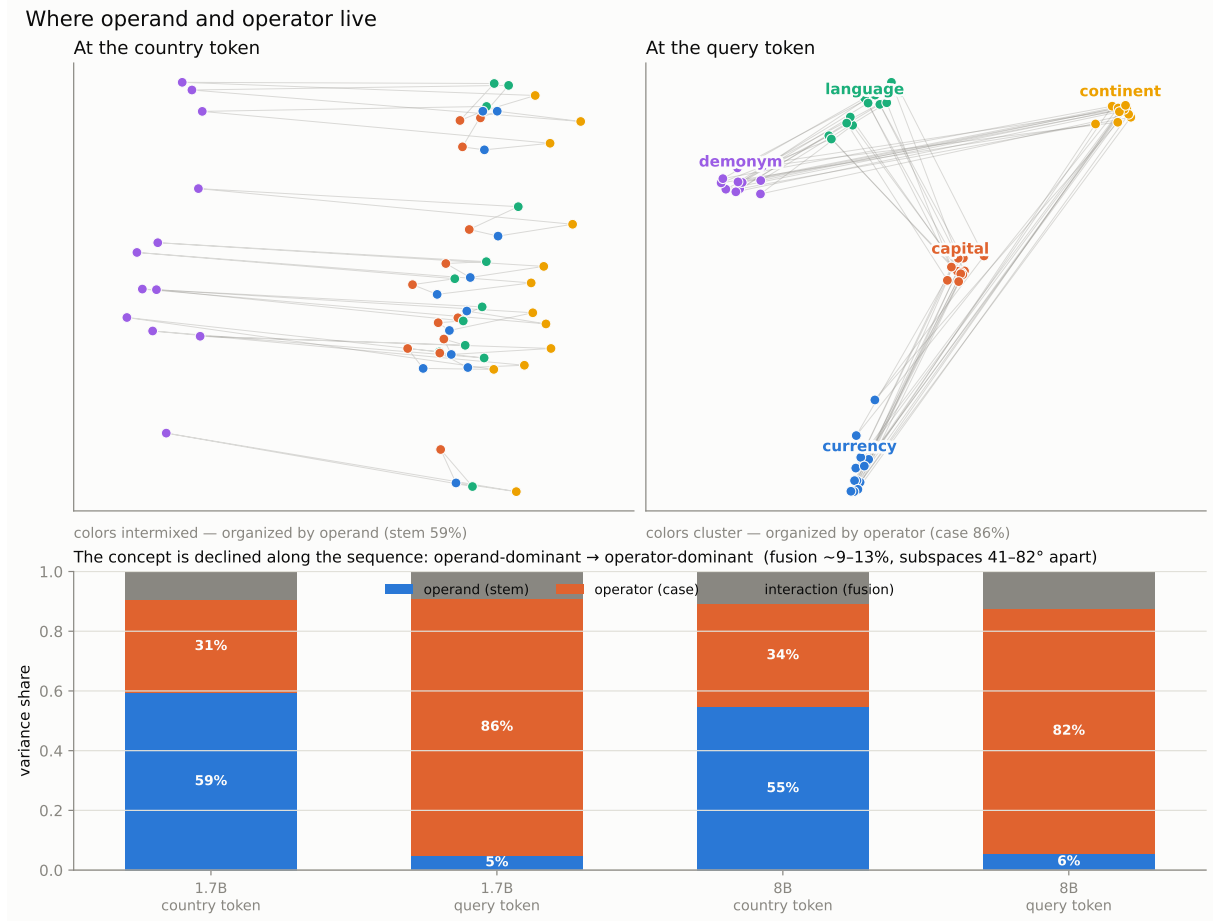


Figure 4: PCA of the 60 workspace vectors $H[\text{operand}, \text{operator}]$, colored by operator. At the country token the cases intermix (operand-organized, stem 59%); at the query token they separate into clean case clusters (operator-organized, case 86%). The faint web links each country's five case-forms — short at the country token, splayed at the query token. Bottom: the variance split, 1.7B and 8B.

on the causal factorization, not on a J-lens readout advantage.

6 Conclusion

An LLM represents a relational operation as a low-dimensional, causally-manipulable operator direction that factorizes from the operand, generalizes to unseen operands, and separates the operation from the word that realizes it — a structure well described as **declension**. This factorization is not universal: it is clean for relational retrieval and breaks down for arithmetic and logic, whose operators are entangled with their operands. The structure is causal; it is not a more-readable subspace.

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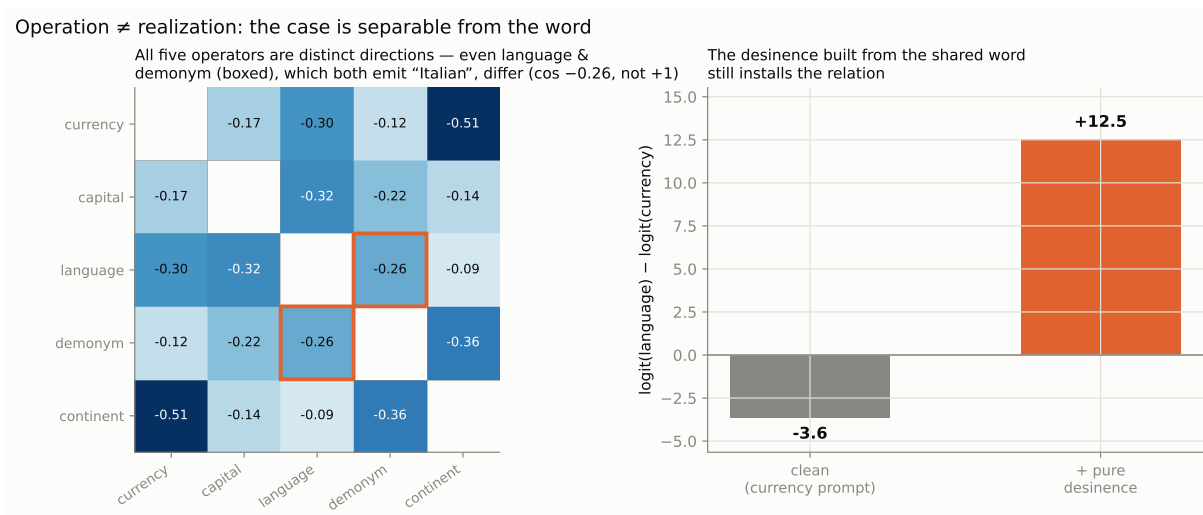


Figure 5: Left: operator-direction cosines — all five operators are distinct directions (no off-diagonal near ± 1); language and demonym, boxed, both emit "Italian" yet differ (cos -0.26). Right: the pure desinence, built where the two share an output word, still installs the relation causally ($-3.6 \rightarrow +12.5$).

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